Context Effects in Attitude Measurement

SurvMeth 632

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What are attitudes?

- Stable <u>dispositions</u>:
  - “a psychological tendency that is expressed by evaluating a particular entity with some degree of favor or disfavor”

Eagley & Chaiken, 1993

- <u>Context sensitive</u>, evaluative judgments:
  - Gives more weight to more recent experience
  - Useful and adaptive for actor, a nuisance for observer (researcher)

Schwarz, 2007

- Actively debated among social psychologists

What are attitudes (2)

- Whatever they are, they can be measured, and this is commonly done in surveys via self-reports
  - Self-reports are the only direct measure of *subjective* phenomena such as attitudes
- Answers to attitude questions are highly influential:
  - opinions about direction of economy can affect stock market, presidential approval can shape policy, market research relies fundamentally on measuring attitudes, e.g., preferences
- They can also be highly sensitive to the <u>measurement context</u>
- How do designers and researchers deal with context sensitivity?

Outline:
Context Effects in Attitude Measurement

- Truth and true value issues when measuring attitudes
- Classic Context Effects
- Forms of context effects
  - Assimilation effects
  - Contrast effects
- Attitude Stability (crystallization)
- Working memory and context effects
- Culture and context
- How automatic are context effects?
- Visual context effects

Attitudes, context and truth

- Reports of attitudes often differ depending on what information has been brought to mind by the context when a respondent reports the attitude
  – Fundamentally different than observable, verifiable behavior for which true value is knowable, at least in principle
  – Because attitudes are context sensitive, respondents can truthfully answer differently in different contexts

Attitudes, context and truth (2)

- Context can be defined by recent conversations, media exposure, environmental stimuli, music, mood, etc.
- In surveys, context includes
  1. previously presented questions, i.e., the *question context*
  - ***temporarily available***, i.e., no longer than time to complete questionnaire
  2. characteristics of respondent
  - ***chronically available*** information, e.g., political ideology
- Are context effects a type of measurement error?

Attitudes, context and truth (3)

- If context leads to unintended <u>interpretation</u> of a question or attitude object, effect can be considered measurement error
  - Question context can affect how current question is interpreted;
  - if this differs from what is intended, the misunderstanding is clearly measurement error
  - e.g., If "How much do you favor or oppose limiting the amount of red meat in your diet?" is preceded by "In how many meals each week do you eat meat of any kind?" may lead *R*s to <u>misinterpret</u> "red meat" as "meat of any kind"

Attitudes, context and truth (4)

- But because attitudes can differ in different contexts, true value can change depending on context
- a recent conversation – or a recent survey question -- may activate stored information or introduce new information
  - e.g., one's (actual) attitudes about a political candidate may be changed by being asked the previous question if it concerns
  1. scandals in which they were involved
  2. popular legislation they championed

Classic Question Order effects: Communist Reporter Question

Do you think the United States should let Communist reporters from other countries come in here and send back to their papers the news as they see it?

Do you think a Communist country like Russia should let American newspaper reporters come in and send back to their papers the news as they see it?

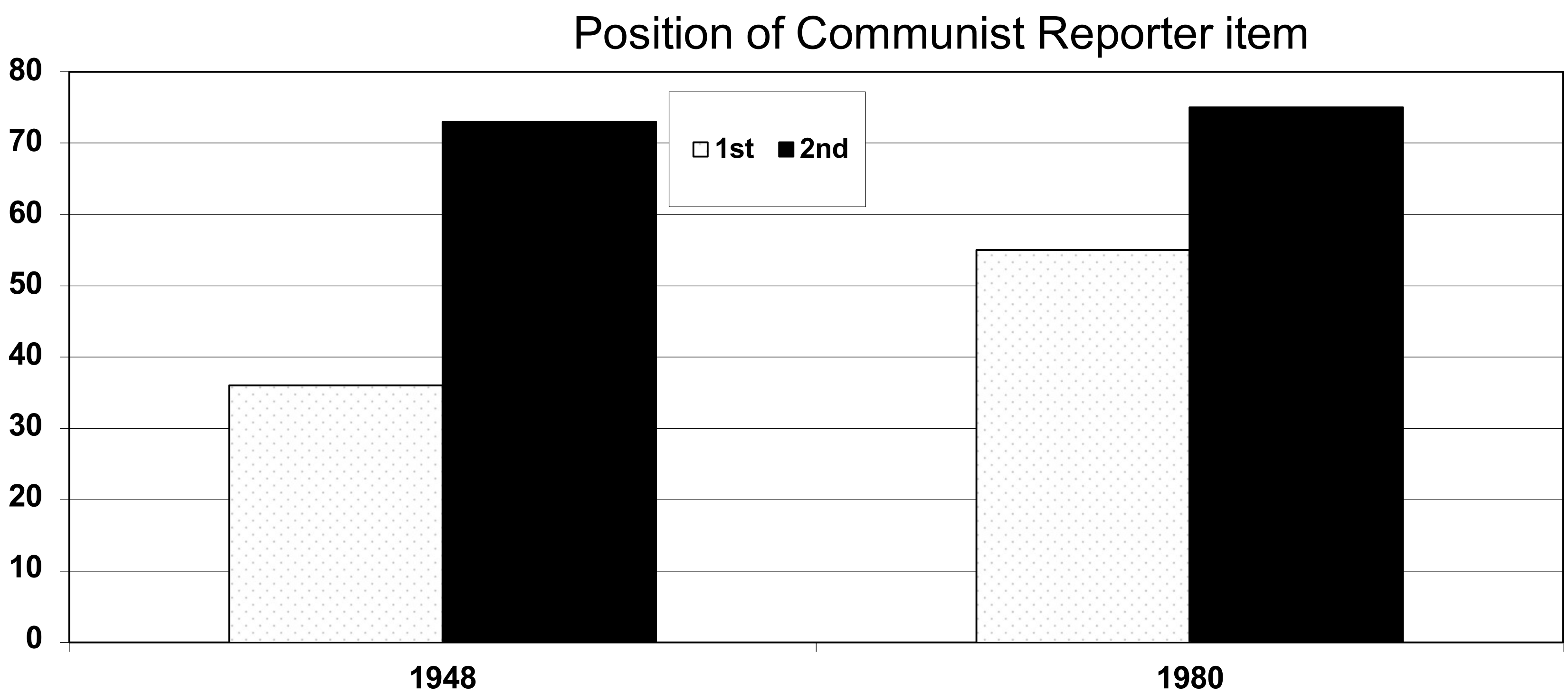
- Communist reporter item supported more in 2nd than 1st position
- First supporting US journalists' right to report from Russia invokes norm of reciprocity – only fair to apply same standard

Communist Reporter (2)

- Schuman ('92) demonstrated that if measuring attitudes over time; must track attitudes in both positions to disentangle effects of norm from current social context (societal change in attitude)
- Amount of support in 2nd position (75%) is <u>constant</u> over 32 years
  - Impact of norm of reciprocity has not changed over time
- Support in 1st position – *when norm is not invoked* – is always lower but <u>increases</u> over same time (36%-55%)
  - reflects drop in fear of Communism

Context Effects on Communist Reporter Question

Schuman, 1992



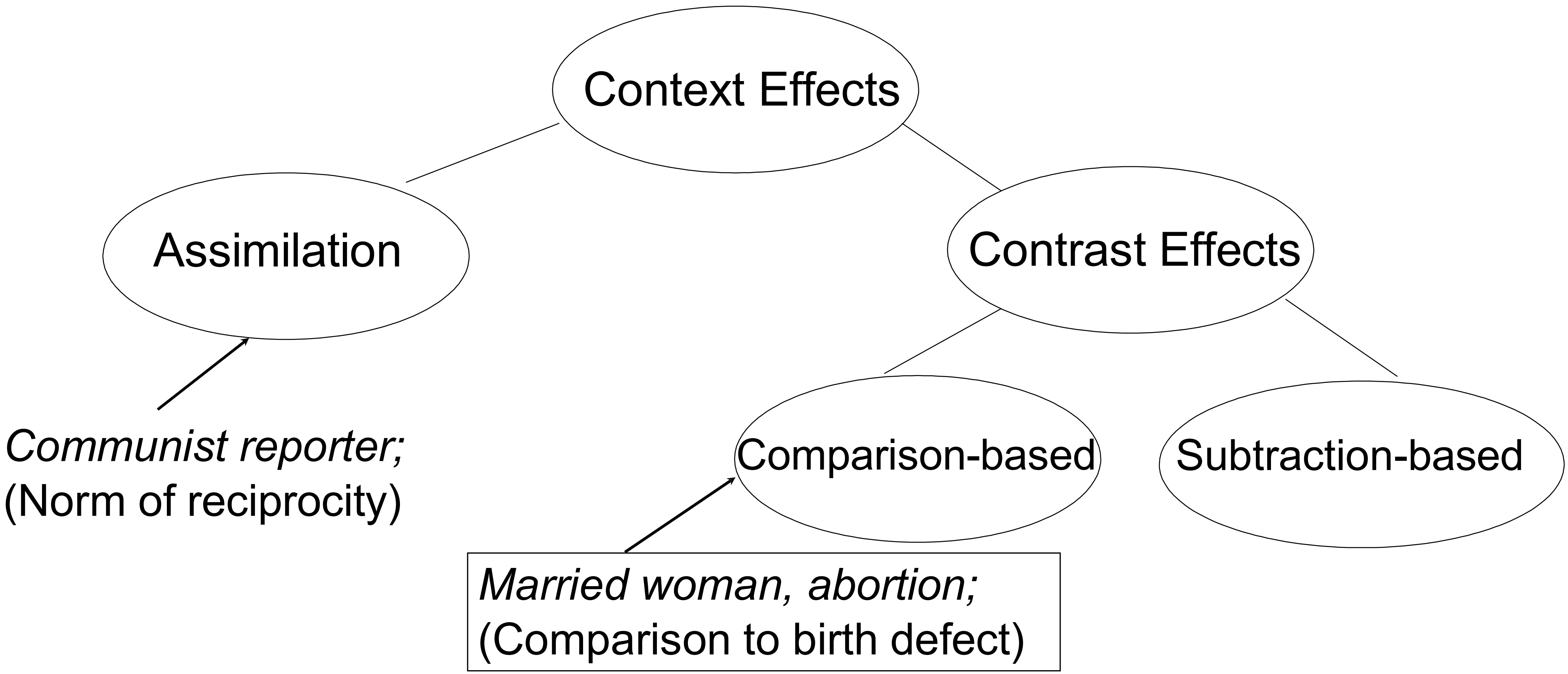
% respondents answering “yes” to communist reporters question

Classic Question Order Effects: Abortion Question

Do you support legal abortions when the woman is married and does not want any more children?

Do you support legal abortions when there is a strong chance of a serious defect in the baby?

- Schuman, Presser, & Ludwig, 1981:
  - married woman 1st: 60.7% (yes to married item)
  - serious defect 1st: 48.1% (yes to married item)
- when married woman item asked first response based on numerous (and unknown) considerations but when asked second, compared to abortion in case of serious defect and thus less compelling


Context Effects
Assimilation
Contrast Effects
Comparison-based
Subtraction-based
Communist reporter;
(Norm of reciprocity)
Married woman, abortion;
(Comparison to birth defect)

Assimilation Effects

- Respondents include info in representation of target (attitude object) that are already in mind from earlier question – *a priming effect*
- If information brought to mind by previous question is used to answer current question, responses should be correlated
  - e.g., Schwarz, Strack & Mai ('91) specific-general condition
- Smaller effect when multiple preceding questions
  - e.g., Schwarz, Strack and Mai ('91) preceded general question by 3 specific questions (work, leisure and marriage)
  - Suggests that > 1 consideration reduces consistency of temporarily available information

Assimilation Effects
Schwarz, Strack and Mai (1991)

	Number of specific questions	
Condition	**One**	**Three**
General-specific		.32
Specific-general	.67*	.46*
Specific-general, with joint lead-in	.18	.48*
Specific-general, explicit inclusion	.61*	.53*
Specific-general, explicit exclusion	.20	.11

r (overall happiness and marital satisfaction)

* *p < .05*

Contrast Effects

- Certain conditions lead to exclusion of temporarily available information from representation of target, eliminating assimilation effect
- Information can be *subtracted* from representation of target or included in standard to which target is *compared*
  - *subtraction-based* effects apply only to that one target
  - *comparison-based* effects for any targets that can be contrasted to standard

Subtraction-based Contrast Effects
Schwarz, Strack and Mai (1991)

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Subtraction-based Contrast Effect

Schwarz & Bless (1992)

Do you happen to know which party Richard von Weizsäcker has been a member of for more than 20 years?

Do you happen to know which office Richard von Weizsäcker holds, setting him apart from all party politics?

Subtraction-based Contrast effect

Assimilation effect

Preceding Question about Richard von Weizsäcker			
	Relating to party membership	No Question	Relating to Presidency
Christian Democrats	$\boxed{6.5}$	$\boxed{5.2}$	$\boxed{3.4}$
Social Democrats	6.3	6.3	6.2

Evaluation of political parties

Comparison-based Contrast Effects

- Context must match evaluative dimension to trigger comparison
- Schwarz, Münkel, and Hippler (‘90) produce or eliminate comparison-based contrast effect by matching or mismatching evaluative dimension
  - asked *R*s to rate typicality of wine, milk and coffee as German drinks
  - Preceded by rating task with *Q* about either (1) frequency of beer- or vodka-drinking in Germany, or (2) caloric content of beer or vodka
    - *Frequency* taps same evaluative dimension as *typicality,* but *caloric conten*t does not
- Estimating frequency leads to comparison-based contrast effect but no effect after estimating caloric content

If evaluative dimension is same in context and target *Q*, comparison (contrast) is possible

	Context Stimulus	
	Vodka	Beer
Preceding question		
Frequency	5.4	4.4
Caloric Content	4.4	4.5

Evaluations of German typicality of wine, milk and coffee

Schwarz, Münkel, and Hippler (‘90)

Attitude Accessibility and Context Effects

- Bassili & Fletcher ('91) measure response time (RT) to "value" questions:
  1. Do you think large companies should have quotas to ensure that a fixed percentage of women are hired, or should women get no special treatment?
  2. a) If answered "should have quotas," asked "Would you feel the same if this meant not hiring the best person?"

     b) If answered "no special treatment", asked "Would you feel the same if this meant that women would remain economically unequal?"
- Those who changed their minds are "movers;" those who did not change their minds are "non-movers"
- Faster Response Times for non-movers (1.744) than movers (2.779)
- Suggests non-movers hold "crystallized," i.e., highly accessible, pre-formed attitudes

Mental Resources and Context Effects

- To include or exclude information in representation of attitude object, it must be available, albeit temporarily
  - temporary maintenance requires mental resources
  - usually assumed to be working memory (WM)
- Individual differences in WM capacity should affect size of question order effects
  - one factor that is associated with degradation of WM capacity is <u>age</u>
- Knäuper (1998) showed that older respondents show smaller context effects than younger respondents
  - Suggests older respondents don't remember how answered earlier *questions* as well as younger respondents

Working Memory, Age, and Context Effects

- In reanalysis of Schuman & Presser (1981), Knäuper (1998) found assimilation effect among younger respondents
  - S&P collected “subjective social class” ratings
  - ½ *R*s previously asked questions about “objective social class,” based on education and occupation, and other ½ asked afterward
  - expected higher association between objective and subjective social class when objective questions asked first
    - assimilation effect
  - In fact, S&P found no effect of previous questions
- Knäuper found assimilation effect for young (18- 54) respondents but not for older (> 55) ones

Direct measurement of Working Memory

- Knauper et al. (2007) asked younger (< 30) and older (70 < M < 80) respondents questions about abortion and labor
- Assessed working memory with reading span task
  - Respondents were read sets of sentences out loud then asked question about content and to write down last word
  - Presented increasingly longer sets of sentences until made error
- Divided older respondents into quartiles on basis working memory
  - Performance on working memory task of top quartile similar to young respondents
- Strong order effect for young respondents and older respondents in top 2 quartiles
- No question order effect for older respondents in lower 2 quartiles, i.e., with smallest WM capacity

Buffer Items and Context Effects

- Questionnaire designers often separate general and specific items with unrelated (buffer) items to reduce context effects
- Ottati et al. (1989) asked respondents if KKK should have freedom of speech then asked general question about freedom of speech
  - Respondents seemed to treat both questions as part of same context, interpreting general question as being about other people, i.e., excluding KKK and producing contrast effect
  - When several unrelated buffer items intervened, relatedness of KKK and general questions seemed to be disrupted, producing assimilation effect
- However, Bishop (1978) found respondents' inability to answer knowledge question about their congressional representative affected their judgments about their political knowledge, even if separated by 101 buffer items

Impact of Culture on Context Effects

- Non-redundancy trigger for context effect requires sensitivity to what is in common ground (working memory for recent discourse)
- Some cultures may place greater value on attention to *others* including content of discourse and, therefore, to what is in common ground
- East Asian culture considered more attuned to others – *interdependent** – than Western culture which is generally oriented around the *individual*
- East Asians should, thus, be more sensitive to what is in common ground, even if not explicitly mentioned, and thus more likely to avoid/reduce redundancy

*also called “collectivist”

Impact of Culture on Context Effects (2)

	Germany	China
Academic-Life	.78	.36
Life-Academic	.53	.50

Correlation between academic and life satisfaction

Haberstroh, et al. (2002)

Impact of Culture on Context Effects (3)

- In more direct test, Haberstroh et al. (2002) briefly induced some German respondents to adopt *interdependent* perspective (like Chinese)
  - *Interdependent* group asked to circle all instances of "we" in short paragraph
  - *Independent* group asked to circle all instances of "I"
- Then asked Academic satisfaction followed by Life satisfaction questions and computed correlations

Independent (I) group:	.76
Interdependent (We) group:	.34

How Automatic are Context Effects?

- Do context effects emerge automatically, or do they require active thought by respondents?
- Non-redundancy (Gricean) trigger for contrast effects could require respondents to *realize* that questions do not ask for same info, *i.e.*, they need to *think* about their answer, so subtraction is not automatic but relatively deliberate
- McCabe & Brannon (2004) tested how automatically contrast effects emerge by comparing correlations for specific-general questions for respondents high and low in Need for Cognition participants
  - Need for cognition tests people's enjoyment of thinking
  - Participants answered specific then general questions after either receiving Joint Lead- in (JL) or not, and then completed need for cognition questionnaire
  - if non-redundancy trigger is automatic, should see contrast effect for all JL participants, irrespective of need for cognition level, but if requires thought, should only observe effect for respondents high in need for cognition

Individual Differences in Need for Cognition and Context Effects

	Joint Lead-In	No Joint Lead-in
High Need for Cognition	$\boxed{.09}$	.40*
Low Need for Cognition	.48*	.45*

*$p < .01$

McCabe & Brannon, 2004

- Joint lead-in only produces contrast effect (i.e., non-significant correlation) for respondents who like to think; suggests requires active thought, i.e., not automatic

Visual Context Effects

- Images are easy and inexpensive to insert into web questionnaires
- Designers frequently use images, presumably to make the page more attractive and, ultimately, increase attention and completion rates
- But do they affect responses and if so, how?
- Couper et al. (2007) explored effect of visual context on responses
  - Visual analogue to question order effects

Visual Context Effects (2)

Questions about this survey?
Email us at life@msiresearch.com
or call toll free 1.866.674.3375

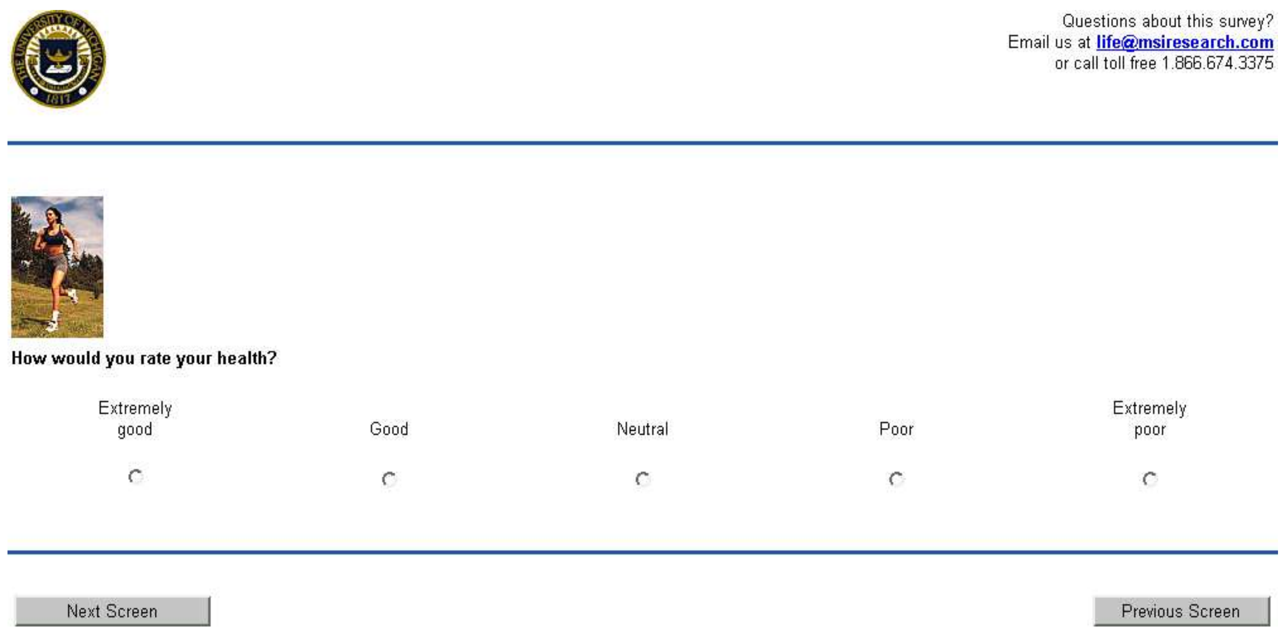
How would you rate your health?

Extremely good	Good	Neutral	Poor	Extremely poor
○	○	○	○	○

Next Screen — Previous Screen

Sick woman in header

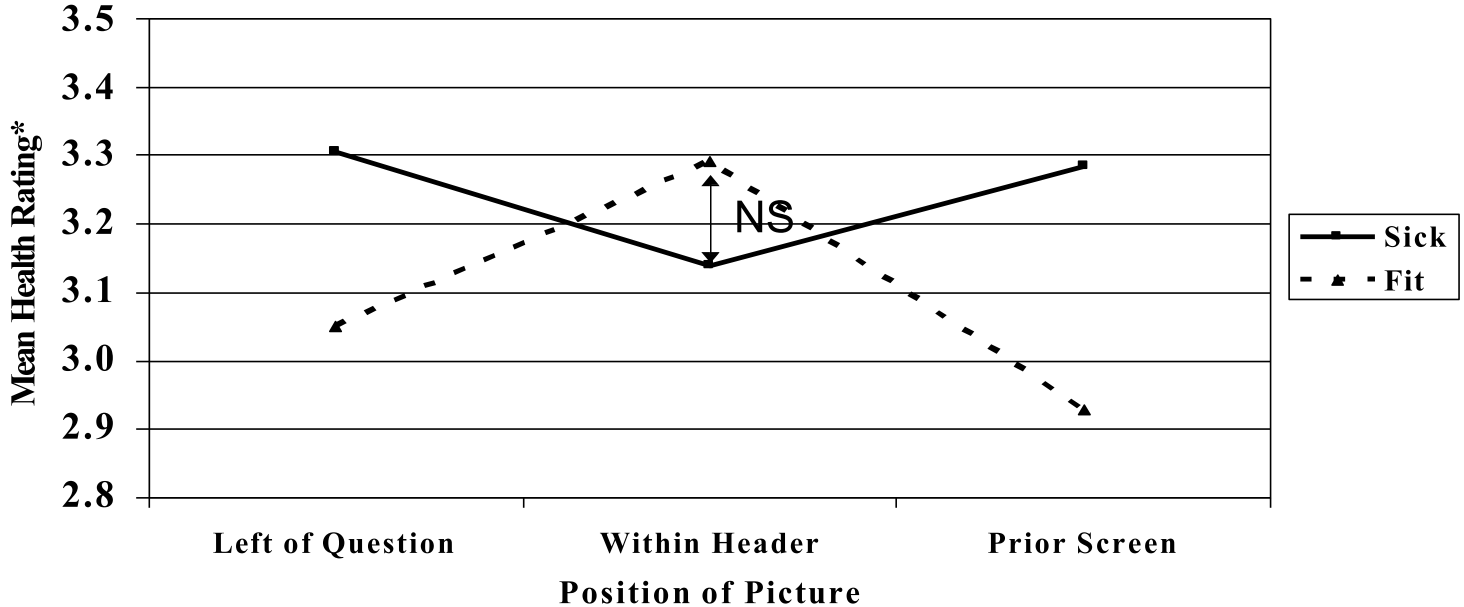
Visual Context Effects (3)



Healthy woman above question

Visual Context Effects (4)

Figure 1: Health Ratings, by Picture and Condition



* Scale range of 1-5; a high score indicates good health

Context Effects: Summary

- Stretches concepts of true value and measurement error
- Assimilation versus contrast effects
  - Reuse of recently activated information vs (1) exclusion/subtraction from target or (2) incorporation into standard of comparison
- Assimilation moderated by working memory capacity (age), sensitivity to conversational context (varies across cultures)
- Contrast effects require thinking, i.e., not automatic
- Visual survey context can operate much like verbal context